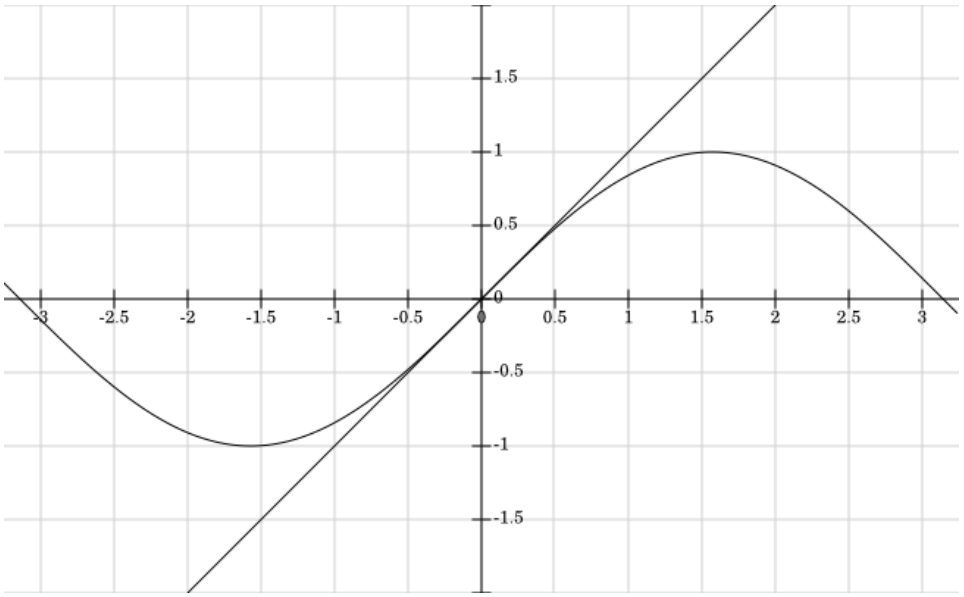
# Sec 8.8 Approximating Functions with Taylor polynomials




$$T_N = \sum_{n=0}^{N} \frac{f^n(a)}{n!}(x-a)^n$$

http://fooplot.com/
#W3sidHlwZSI6MCwiZXEiOiJzaW4oeCkiLCJjb2xvciI6IiMwMDAwMDAifSx7InR5cGUiOjAsImVxIjoieCIsImNvbG9yIjoiIzAwMDAwMCJ9LHsidHlwZSI6MCwiZXEiOiIiLCJjb2xvciI6IiMwMDAwMDAifSx7InR5cGUiOjEwMDAsIndpbmRvdyI6WyItMy4yNSIsIjMuMjUiLCItMiIsIjIiXX1d

# Sec 8.8 Approximating Functions with Taylor polynomials

How can we estimate how far we are off from the actual function?

**9 Taylor's Inequality** If $|f^{(n+1)}(x)| \leq M$ for $|x - a| \leq d$, then the remainder $R_n(x)$ of the Taylor series satisfies the inequality

$$|R_n(x)| \leq \frac{M}{(n+1)!}|x - a|^{n+1} \qquad \text{for } |x - a| \leq d$$

Ex Estimate the error of the 3rd degree Taylor polynomial for sin(x) centered at 0 on the interval defined by |x|<1 What about |x|<20?

# Sec 8.8 Approximating Functions with Taylor polynomials

**Alternating Series Estimation Theorem** If $s = \Sigma (-1)^{n-1} b_n$ is the sum of an alternating series that satisfies

$$\text{(i) } b_{n+1} \leq b_n \qquad \text{and} \qquad \text{(ii) } \lim_{n \to \infty} b_n = 0$$

then

$$|R_n| = |s - s_n| \leq b_{n+1}$$

How far off is the third degree taylor polynomial centered at 0 for sin(x) at x=1/2 what about x=20?

# Sec 8.8 Approximating Functions with Taylor polynomials

HW Example Approximate the Taylor polynomial for $f(x)=x\sin(x)$ at $x=a$ with $n=4$ Estimate the accuracy when $-1<x<1$